

2a. $J(w) = \|H_w(x), y\|^2$ or $J(w) = \overbrace{y \cdot -\ln(H_w(x))}^{\text{cost if } y = 1} - \overbrace{((1-y) \cdot -\ln(1-H_w(x)))}^{\text{cost if } y = 0}$

2b. $H_w(x)$ is the predicted probability value of x , y is the actual probability value, $J(w)$ is the objective function to be minimized. N is the number of samples and D is the number of the original features.

$w \in \mathbb{R}^{D+1}, \text{ space } x \in \mathbb{R}^{N \times D}, y \in \mathbb{R}^{N \times 1}$

2c. No

2d. Supervised

2e. Classification

3. The weights can grow unbounded, ergo overfit to the data and become unstable and has no finite optimal solution.

4. By adding the square of the weights, large weights score worse. This forces the optimal solution to balance accuracy prediction and weight size.

8.

23-24 Part 4

a Logistic regression doesn't assume Gaussian distributions,
logistic regression doesn't have an analytical solution

b w : the weights/model parameters

x : the inputted data point

y : the true label of x

$h_w(x)$: the predicted label of x

$p(y=1|x;w)$: probability (given ~~the~~ some weights)
that x belongs to label 1

c Because the output of logistic regression ~~is~~ relates additively
to its inputs/parameters

$$d \frac{1}{1 + \exp(2\ln(3) - \ln(3)x)} = 0.25$$

$$1 + \exp(2\ln(3) - \ln(3)x) = 4 \Rightarrow \exp(2\ln(3) - \ln(3)x) = 3$$

$$\ln(3)(2-x) = \ln(3) \Rightarrow 2-x=1 \Rightarrow x=1$$

e No, because a line from $J(1)$ to $J(4)$ crosses $J(w)$ in the middle
ie. it has a maximum